

HALL BEACH, NU, Canada

WMO: 713200

Lat: **68.780N**Lon: **81.240W**Elev: **8**StdP: **101.23**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-43.0	-41.4	-45.5	0.0	-41.8	-44.9	0.0	-41.1	14.4	-26.4	13.3	-27.9	4.7	310	1.128

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.9	14.3	11.1	12.0	9.4	10.1	8.1	11.2	13.8	9.6	11.7	8.2	10.0	4.4	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.5	7.4	12.8	8.0	6.6	10.9	6.7	6.1	9.3	32.3	14.0	28.4	11.8	25.3	10.1	17.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	12.4	11.1	DB	-44.5	19.3	2.4	2.2	-46.3	20.9	-47.7	22.2	-49.0	23.4	-50.7	25.1	(4)
(5)			WB	-44.2	14.2	2.0	1.3	-45.6	15.1	-46.7	15.8	-47.9	16.5	-49.3	17.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-12.6	-30.6	-31.5	-27.8	-18.5	-7.8	1.1	6.6	5.5	0.7	-6.9	-17.7	-25.1	(6)
(7)		DBStd	14.74	6.77	6.62	6.77	6.33	5.04	2.93	2.92	2.53	2.78	5.00	7.16	7.41	(7)
(8)		HDD10.0	8244	1258	1161	1173	856	553	266	112	142	278	524	832	1089	(8)
(9)		HDD18.3	11275	1517	1394	1431	1106	811	516	363	398	528	782	1082	1348	(9)
(10)		CDD10.0	11	0	0	0	0	0	1	8	3	0	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.8	5.8	6.0	6.0	5.7	5.5	5.1	4.7	5.2	6.0	6.4	6.5	6.0	(14)
(15)	Precipitation	PrecAvg	224	7	6	11	12	16	19	29	46	29	23	16	10	(15)
(16)		PrecMax	402	24	24	50	35	48	80	85	159	82	55	52	32	(16)
(17)		PrecMin	88	0	0	0	0	2	0	3	3	2	3	0	1	(17)
(18)		PrecStd	73	6	6	12	9	11	15	19	31	20	13	12	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-5.9	-9.8	-6.9	-0.9	1.5	14.0	18.3	17.0	9.3	1.6	-1.1	-3.4	(19)
(20)			MCWB	-6.1	-10.0	-7.3	-1.2	1.0	10.6	13.2	12.9	7.0	1.0	-1.4	-3.6	(20)
(21)		2%	DB	-13.4	-14.5	-11.4	-3.7	0.6	9.3	15.2	12.9	6.8	0.6	-2.3	-7.4	(21)
(22)			MCWB	-13.8	-14.8	-11.8	-4.1	0.1	7.4	11.6	10.1	5.3	0.2	-2.6	-7.7	(22)
(23)		5%	DB	-16.6	-17.5	-14.0	-6.0	-0.1	6.6	13.1	10.8	5.4	0.0	-4.0	-10.8	(23)
(24)			MCWB	-17.0	-17.8	-14.5	-6.6	-0.5	5.1	10.2	8.7	4.2	-0.3	-4.5	-11.1	(24)
(25)		10%	DB	-20.1	-20.8	-16.6	-8.2	-0.9	4.8	11.3	9.3	4.1	-0.6	-6.2	-13.5	(25)
(26)			MCWB	-20.5	-21.2	-17.1	-8.7	-1.3	3.7	9.0	7.5	3.4	-1.0	-6.5	-13.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.0	-10.0	-7.2	-1.2	1.2	10.8	13.7	13.3	7.3	1.2	-1.4	-3.6	(27)
(28)			MCDWB	-5.9	-9.9	-6.9	-1.0	1.3	13.5	17.2	16.6	8.8	1.3	-1.2	-3.5	(28)
(29)		2%	WB	-13.7	-14.7	-11.8	-4.1	0.2	7.6	12.0	10.5	5.6	0.4	-2.6	-7.7	(29)
(30)			MCDWB	-13.4	-14.3	-11.5	-3.8	0.4	9.3	14.9	12.4	6.7	0.5	-2.3	-7.4	(30)
(31)		5%	WB	-17.0	-17.8	-14.4	-6.6	-0.4	5.2	10.4	8.8	4.3	-0.2	-4.4	-11.1	(31)
(32)			MCDWB	-16.6	-17.4	-14.0	-6.1	-0.2	6.6	12.9	10.6	5.1	0.0	-4.0	-10.9	(32)
(33)		10%	WB	-20.4	-21.1	-17.0	-8.7	-1.3	3.7	9.1	7.6	3.5	-0.9	-6.5	-13.9	(33)
(34)			MCDWB	-20.0	-20.8	-16.6	-8.2	-0.9	4.7	11.3	9.1	4.1	-0.7	-6.2	-13.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.2	7.3	8.5	9.1	6.7	3.7	5.9	4.7	3.2	5.3	7.6	7.3	(35)
(36)			MCDBR	9.8	10.5	10.5	8.5	4.7	6.7	9.3	7.8	5.2	3.5	7.3	8.4	(36)
(37)			MCWBR	9.8	10.5	10.3	8.3	4.6	5.0	6.5	5.5	4.0	3.6	7.4	8.4	(37)
(38)		5% WB	MCDBR	9.7	10.5	10.5	8.4	4.5	6.6	9.3	7.5	4.9	3.4	7.3	8.5	(38)
(39)			MCWBR	9.7	10.5	10.3	8.3	4.5	5.1	6.9	5.6	3.9	3.6	7.5	8.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.171	0.204	0.249	0.322	0.323	0.322	0.355	0.326	0.292	0.238	0.159	N/A	(40)	
(41)		taud	1.840	2.046	2.158	1.974	2.134	2.303	2.354	2.516	2.516	2.178	1.916	N/A	(41)	
(42)		Ebn at Noon	61	595	789	799	857	873	826	820	758	531	67	0	(42)	
(43)		Edh at Noon	6	44	77	127	123	107	99	76	57	38	4	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.02	0.47	2.15	4.50	6.07	6.46	5.41	3.59	1.79	0.52	0.07	0.00	(44)	
(45)		RadStd	0.00	0.05	0.08	0.15	0.36	0.41	0.48	0.24	0.20	0.06	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page